Hospital Analytics Using SQL

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Project Objective

Analyze hospital encounter data to identify trends in patient admissions, encounter patterns, treatment costs, insurance coverage, and readmission behavior using SQL.

Dataset Includes:

- Patients table
- Encounters table
- Procedures table
- Payers table

How many total encounters occurred each year?

```sql
SELECT
    YEAR(START) AS Year,
    COUNT(DISTINCT id) AS Total_Encounters
FROM
    encounters
GROUP BY Year
ORDER BY Year;
```

OUTPUT :

Year	Total_Encount...
2011	1336
2012	2106
2013	2495
2014	3885
2015	2469
2016	2451
2017	2360
2018	2292
2019	2228
2020	2519
2021	3530
2022	220

For each year, what percentage of all encounters belonged to each encounter class?

```
SELECT
    YEAR(START) AS Year,
    ENCOUNTERCLASS,
    COUNT(DISTINCT id) AS Total_Encounters,
    COUNT(DISTINCT id) / (SELECT
            COUNT(*)
        FROM
            encounters
        WHERE
            YEAR(START) = Year) * 100 AS encounter_class_percent
FROM
    encounters
GROUP BY Year , ENCOUNTERCLASS
ORDER BY Year;
```

OUTPUT :

Year	ENCOUNTERCLASS	Total_Encount...	encounter_class_perc
2011	ambulatory	667	49.9300
2011	emergency	55	4.1200
2011	inpatient	83	6.2100
2011	outpatient	327	24.4800
2011	urgentcare	30	2.2500
2011	wellness	174	13.0200
2012	ambulatory	895	42.5000
2012	emergency	183	8.6900
2012	inpatient	93	4.4200
2012	outpatient	444	21.0800
2012	urgentcare	299	14.2000
2012	wellness	192	9.1200
2013	ambulatory	1106	44.3300
2013	emergency	225	9.0200
2013	inpatient	134	5.3700
2013	outpatient	485	19.4400
2013	urgentcare	359	14.3900
2013	wellness	186	7.4500
2014	ambulatory	2341	60.2600

What percentage of encounters were over 24 hours versus under 24 hours?

```
with t_dur as (select CASE
when timediff(STOP,START)>'24:00:00' then "Over 24 hours"
else "Under 24 hrs" end AS duration from encounters)
select duration ,
count(*)/(select count(*) from encounters)*100 as percent ,
count(*) AS ENCOUNTERS
from t_dur
group by duration;
```

OUTPUT :

duration	percent	ENCOUNTERS
Under 24 hrs	99.7800	27830
Over 24 hours	0.2200	61

How many encounters had zero payer coverage, and what percentage of total encounters does this represent?

```sql
SELECT
    COUNT(DISTINCT id) AS zero_payer_coverage,
    ROUND(COUNT(DISTINCT id) /
    (SELECT COUNT(*) FROM encounters) * 100,1)
    AS percentage
FROM
    encounters
WHERE
    PAYER_COVERAGE = 0;
```

OUTPUT :

zero_payer_covera...	percentage
13586	48.7

What are the top 10 most frequent procedures performed and the average base cost for each?

```
SELECT
    DESCRIPTION,
    COUNT(*) AS procedure_frequency,
    ROUND(AVG(BASE_COST), 1) AS average_base_cost
FROM
    procedures
GROUP BY DESCRIPTION
ORDER BY procedure_frequency DESC
LIMIT 10;
```

OUTPUT :

DESCRIPTION	procedure_freque...	average_base_cost
Assessment of health and social care needs (pr...	4596	431.0
Hospice care (regime/therapy)	4098	431.0
Depression screening (procedure)	3614	431.0
Depression screening using Patient Health Que...	3614	431.0
Assessment of substance use (procedure)	2906	431.0
Renal dialysis (procedure)	2746	1004.1
Assessment using Morse Fall Scale (procedure)	2422	431.0
Assessment of anxiety (procedure)	2288	431.0
Medication Reconciliation (procedure)	2284	509.1
Screening for drug abuse (procedure)	1484	431.0

What are the top 10 procedures with the highest average base cost and the number of times they were performed?

```
SELECT
    DESCRIPTION,
    ROUND(AVG(BASE_COST), 1) AS average_base_cost ,  COUNT(*) AS procedure_frequency
FROM
    procedures
GROUP BY DESCRIPTION
ORDER BY average_base_cost DESC
LIMIT 10;
```

OUTPUT :

DESCRIPTION	average_base_cost	procedure_freque...
Admit to ICU (procedure)	206260.4	5
Coronary artery bypass grafting	47085.9	9
Lumpectomy of breast (procedure)	29353.0	5
Hemodialysis (procedure)	29299.6	27
Insertion of biventricular implantable cardioverte...	27201.0	4
Electrical cardioversion	25903.1	1383
Partial resection of colon	25229.3	7
Fine needle aspiration biopsy of lung (procedure)	23141.0	1
Percutaneous mechanical thrombectomy of port...	20228.0	57
Percutaneous coronary intervention	19728.0	9

What is the average total claim cost for encounters, broken down by payer?

```
SELECT
    p.name,
    ROUND(AVG(e.TOTAL_CLAIM_COST), 1) AS average_total_claim_cost
FROM
    encounters e
        JOIN
    payers p ON e.PAYER = p.ID
GROUP BY e.PAYER;
```

OUTPUT :

name	average_total_claim_c.
Blue Cross Blue Shield	3245.6
NO_INSURANCE	5593.2
Dual Eligible	1696.2
Medicare	2167.6
Cigna Health	2997.0
Humana	3269.3
Medicaid	6205.2
Aetna	2767.0
Anthem	4236.8
UnitedHealthcare	2848.3

How many unique patients were admitted each quarter over time?

```
SELECT
    YEAR(START) as Year,
    QUARTER(START) as Quarter,
    COUNT(DISTINCT id) AS number_of_patients
FROM
    encounters
GROUP BY YEAR(START) , QUARTER(START);
```

OUTPUT :

Year	Quarter	number_of_patie..
2011	1	301
2011	2	375
2011	3	363
2011	4	297
2012	1	469
2012	2	570
2012	3	534
2012	4	533
2013	1	568
2013	2	711
2013	3	607
2013	4	609
2014	1	1845
2014	2	693

Which patients had the most readmissions?

```sql
SELECT
   e.PATIENT, CONCAT( p.FIRST ," ", p.LAST) as Name, count(*) as num
FROM
    encounters e join patients p on e.PATIENT= p.Id
GROUP BY e.PATIENT order by num desc limit 10 ;
```

OUTPUT :

PATIENT	Name	num
1712d26d-822d-1e3a-2267-0a9dba31d7c8	Kimberly627 Collier206	1381
5e055638-0dad-dfd5-005d-1e74b6fd29ac	Shani239 Parisian75	887
3de74169-7f67-9304-91d4-757e0f3a14d2	Mariano761 OKon634	877
3f523789-55f3-bb31-2757-4803ca6a9c2a	Gail741 Glover433	447
5dcb295d-92df-a147-ebcc-aa49b6262830	Ward668 Nicolas769	441
442dc617-c7f2-0513-15cd-c35c4efdba73	Marcos263 Parker433	392
b4ab9ab3-f52a-751e-a990-3bd5654d5870	Mariano761 Schroeder447	383
9b8ae606-5059-b3a6-19c7-c812901898bb	Song837 Heaney114	364
b4671e80-7e87-9260-ca31-6a9397d465d1	Issac619 Predovic534	296
ff331e5c-ab16-e218-f39a-63e11de1ed75	Eugene421 Abernathy524	291

Key Insights

📌Hospital encounters increased from 1,336 (2011) to 3,885 (2014), showing growing patient inflow.

📌 Ambulatory visits dominated hospital encounters, making up the majority of patient visits over the years.

📌 99.78% of hospital stays lasted under 24 hours, showing most treatments were short-term.

📌 Around 48.7% of encounters had zero payer coverage, indicating many patients may have been self-paying.

📌Health assessments and therapy-related procedures were the most commonly performed treatments.

📌 ICU admissions and heart surgeries had the highest treatment costs despite being performed less frequently.

Thank You!

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SQL | Excel | Tableau | Power BI |
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